DÉMO · CHANGEMENTS ENTRE LES ÉTAPES

Redux puis Redux Toolkit

Chaque étape comparée à la précédente : ce qui est ajouté et ce qui est retiré.

`+ ligne ajoutée` `– ligne retirée` `@@ position dans le fichier`

1. **Étape 1 → 2** · etape-2-multi-reducers (2 fichiers)
2. **Étape 2 → 3** · etape-3-provider (7 fichiers)
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Étape 1 → 2 etape-1-basic → etape-2-multi-reducers

Un reducer par produit, combinés avec combineReducers

MODIFIÉ **src/pages/Home.jsx** +11 −10

```
@@ -1,18 +1,19 @@
- import phone from "../assets/phone.png";
- import tv from "../assets/tv.png";
- import camera from "../assets/camera.png";
-
+ import phoneImg from "../assets/phone.png";
+ import tvImg from "../assets/tv.png";
+ import cameraImg from "../assets/camera.png";
  import { buyPhone, buyTV, buyCamera, store } from "../store/index.js";
  function Home() {
    const state = store.getState();
+   const { phone, tv, camera } = state;
+
    return (
      <div>
        <h2>{"Welcome to our Application"}</h2>
        <div className="container">
          <div className="item">
-           <img src={phone} alt="" width={100} height={100} />
+           <img src={phoneImg} alt="" width={100} height={100} />
            <div className="availability" id="phones">
-             {state.phones} available
+             {phone.total} available
            </div>
            <button
              className="button"
@@ -24,9 +25,9 @@
            </button>
          </div>
          <div className="item">
-           <img src={tv} alt="" width={100} height={100} />
+           <img src={tvImg} alt="" width={100} height={100} />
            <div className="availability" id="tvs">
-             {state.tvs} available
+             {tv.total} available
            </div>
            <button
              className="button"
@@ -38,9 +39,9 @@
```

```
38  39            </button>
39  40          </div>
40  41          <div className="item">
41      -         <img src={camera} alt="" width={100} height={100} />
    42  +         <img src={cameraImg} alt="" width={100} height={100} />
42  43            <div className="availability" id="cameras">
43      -           {state.cameras} available
    44  +           {camera.total} available
44  45            </div>
45  46            <button
46  47              className="button"
```

MODIFIÉ **src/store/index.js** +50 −22

```
        @@ -1,4 +1,4 @@
1       - import { createStore } from "redux";
    1   + import { combineReducers, createStore } from "redux";
2   2
3   3   // Action Types
4   4
        @@ -27,45 +27,73 @@
27  27  // reducer
28  28
29  29  // initial state
30      - const initialState = {
31      -   phones: 5,
32      -   tvs: 3,
33      -   cameras: 8,
    30  + const initialStatePhone = {
    31  +   total: 5,
    32  + };
    33  +
    34  + const initialStateTV = {
    35  +   total: 3,
    36  + };
    37  +
    38  + const initialStateCamera = {
    39  +   total: 8,
34  40  };
35  41
36  42  // (previousState, action) => newState
37      - const phoneReducer = (state = initialState, action) => {
    43  + const phoneReducer = (state = initialStatePhone, action) => {
38  44    switch (action.type) {
39  45      case BUY_PHONE:
40  46        return {
41  47          ...state,
42      -         phones: state.phones - 1,
43      -       };
44      -     case BUY_TV:
45      -       return {
46      -         ...state,
47      -         tvs: state.tvs - 1,
48      -       };
49      -     case BUY_CAMERA:
50      -       return {
51      -         ...state,
52      -         cameras: state.cameras - 1,
    48  +         total: state.total - 1,
53  49        };
54  50      default:
55  51        return state;
56  52    }
57  53  };
```

```
58  54
59      - // store
60      - const store = createStore(phoneReducer);
        55  + // (previousState, action) => newState
        56  + const tvsReducer = (state = initialStateTV, action) => {
        57  +   switch (action.type) {
        58  +     case BUY_TV:
        59  +       return {
        60  +         ...state,
        61  +         total: state.total - 1,
        62  +       };
        63  +     default:
        64  +       return state;
        65  +   }
        66  + };
        67  +
        68  + // (previousState, action) => newState
        69  + const camerasReducer = (state = initialStateCamera, action) => {
        70  +   switch (action.type) {
        71  +     case BUY_CAMERA:
        72  +       return {
        73  +         ...state,
        74  +         total: state.total - 1,
        75  +       };
        76  +     default:
        77  +       return state;
        78  +   }
        79  + };
        80  +
        81  + const rootReducer = combineReducers({
        82  +   phone: phoneReducer,
        83  +   tv: tvsReducer,
        84  +   camera: camerasReducer,
        85  + });
        86  +
        87  + const store = createStore(rootReducer);
61  88
62  89    store.subscribe(() => {
63  90      const state = store.getState();
        91  +   const { phone, tv, camera } = state;
64  92      console.log("Updated State:", state);
65  93      // Update the UI elements directly (Test)
66      -   document.getElementById("phones").innerHTML = `${state.phones} available`;
67      -   document.getElementById("tvs").innerHTML = `${state.tvs} available`;
68      -   document.getElementById("cameras").innerHTML = `${state.cameras} available`;
        94  +   document.getElementById("phones").innerHTML = `${phone.total} available`;
        95  +   document.getElementById("tvs").innerHTML = `${tv.total} available`;
        96  +   document.getElementById("cameras").innerHTML = `${camera.total} available`;
69  97    });
70  98
71  99    export { store, buyPhone, buyTV, buyCamera, phoneReducer };
```

Étape 2 → 3 etape-2-multi-reducers → etape-3-provider

Dossier store/ (actions, reducers, store), <Provider>, useSelector, useDispatch

MODIFIÉ `src/App.jsx` +8 −2

```
@@ -1,8 +1,14 @@
  import "./App.css";
- import Home from "./pages/Home.jsx";
+ import Home from "./pages/Home.tsx";
+ import { Provider } from "react-redux";
+ import { store } from "./store/store.ts";

  function App() {
-   return <Home />;
+   return (
+     <Provider store={store}>
+       <Home />
+     </Provider>
+   );
  }

  export default App;
```

SUPPRIMÉ `src/pages/Home.jsx` +0 −60

```
@@ -1,60 +1,0 @@
- import phoneImg from "../assets/phone.png";
- import tvImg from "../assets/tv.png";
- import cameraImg from "../assets/camera.png";
- import { buyPhone, buyTV, buyCamera, store } from "../store/index.js";
- function Home() {
-   const state = store.getState();
-   const { phone, tv, camera } = state;
-
-   return (
-     <div>
-       <h2>{"Welcome to our Application"}</h2>
-       <div className="container">
-         <div className="item">
-           <img src={phoneImg} alt="" width={100} height={100} />
-           <div className="availability" id="phones">
-             {phone.total} available
-           </div>
-           <button
-             className="button"
-             onClick={() => {
-               store.dispatch(buyPhone());
-             }}
-           >
-             Buy
-           </button>
-         </div>
-         <div className="item">
-           <img src={tvImg} alt="" width={100} height={100} />
-           <div className="availability" id="tvs">
-             {tv.total} available
-           </div>
-           <button
-             className="button"
-             onClick={() => {
-               store.dispatch(buyTV());
-             }}
-           >
```

```
-            Buy
-          </button>
-        </div>
-        <div className="item">
-          <img src={cameraImg} alt="" width={100} height={100} />
-          <div className="availability" id="cameras">
-            {camera.total} available
-          </div>
-          <button
-            className="button"
-            onClick={() => {
-              store.dispatch(buyCamera());
-            }}
-          >
-            Buy
-          </button>
-        </div>
-      </div>
-    </div>
-  );
- }
-
- export default Home;
```

AJOUTÉ `src/pages/Home.tsx` +69 −0

```
@@ -1,0 +1,69 @@
+ // @ts-ignore
+ import phoneImg from "../assets/phone.png";
+ // @ts-ignore
+ import tvImg from "../assets/tv.png";
+ // @ts-ignore
+ import cameraImg from "../assets/camera.png";
+
+ import { store } from "../store/store";
+ import { buyPhone, buyTV, buyCamera } from "../store/actions";
+ import { useSelector, useDispatch } from "react-redux";
+
+ function Home() {
+   const dispatch = useDispatch();
+   const phone = useSelector((state: any) => state.phone);
+   const camera = useSelector((state: any) => state.camera);
+   const tv = useSelector((state: any) => state.tv);
+
+   return (
+     <div>
+       <h2>{"Welcome to our Application"}</h2>
+       <div className="container">
+         <div className="item">
+           <img src={phoneImg} alt="" width={100} height={100} />
+           <div className="availability" id="phones">
+             {phone.total} available
+           </div>
+           <button
+             className="button"
+             onClick={() => {
+               dispatch(buyPhone());
+             }}
+           >
+             Buy
+           </button>
+         </div>
+         <div className="item">
+           <img src={tvImg} alt="" width={100} height={100} />
+           <div className="availability" id="tvs">
```

```
39 +             {tv.total} available
40 +           </div>
41 +           <button
42 +             className="button"
43 +             onClick={() => {
44 +               dispatch(buyTV());
45 +             }}
46 +           >
47 +             Buy
48 +           </button>
49 +         </div>
50 +         <div className="item">
51 +           <img src={cameraImg} alt="" width={100} height={100} />
52 +           <div className="availability" id="cameras">
53 +             {camera.total} available
54 +           </div>
55 +           <button
56 +             className="button"
57 +             onClick={() => {
58 +               dispatch(buyCamera());
59 +             }}
60 +           >
61 +             Buy
62 +           </button>
63 +         </div>
64 +       </div>
65 +     </div>
66 +   );
67 + }
68 +
69 + export default Home;
```

AJOUTÉ **src/store/actions.ts** +25 −0

```
@@ -1,0 +1,25 @@
1  + // Action Types
2  +
3  + const BUY_PHONE = "BUY_PHONE";
4  + const BUY_TV = "BUY_TV";
5  + const BUY_CAMERA = "BUY_CAMERA";
6  +
7  + const buyPhone = () => {
8  +   return {
9  +     type: BUY_PHONE,
10 +   };
11 + };
12 +
13 + const buyTV = () => {
14 +   return {
15 +     type: BUY_TV,
16 +   };
17 + };
18 +
19 + const buyCamera = () => {
20 +   return {
21 +     type: BUY_CAMERA,
22 +   };
23 + };
24 +
25 + export { BUY_PHONE, BUY_TV, BUY_CAMERA, buyPhone, buyTV, buyCamera };
```

SUPPRIMÉ **src/store/index.js** +0 −99

```
@@ -1,99 +1,0 @@
1  - import { combineReducers, createStore } from "redux";
```

```
-
- // Action Types
-
- const BUY_PHONE = "BUY_PHONE";
- const BUY_TV = "BUY_TV";
- const BUY_CAMERA = "BUY_CAMERA";
-
- const buyPhone = () => {
-   return {
-     type: BUY_PHONE,
-   };
- };
-
- const buyTV = () => {
-   return {
-     type: BUY_TV,
-   };
- };
-
- const buyCamera = () => {
-   return {
-     type: BUY_CAMERA,
-   };
- };
-
- // reducer
-
- // initial state
- const initialStatePhone = {
-   total: 5,
- };
-
- const initialStateTV = {
-   total: 3,
- };
-
- const initialStateCamera = {
-   total: 8,
- };
-
- // (previousState, action) => newState
- const phoneReducer = (state = initialStatePhone, action) => {
-   switch (action.type) {
-     case BUY_PHONE:
-       return {
-         ...state,
-         total: state.total - 1,
-       };
-     default:
-       return state;
-   }
- };
-
- // (previousState, action) => newState
- const tvsReducer = (state = initialStateTV, action) => {
-   switch (action.type) {
-     case BUY_TV:
-       return {
-         ...state,
-         total: state.total - 1,
-       };
-     default:
-       return state;
-   }
- };
```

```
-
- // (previousState, action) => newState
- const camerasReducer = (state = initialStateCamera, action) => {
-   switch (action.type) {
-     case BUY_CAMERA:
-       return {
-         ...state,
-         total: state.total - 1,
-       };
-     default:
-       return state;
-   }
- };
-
- const rootReducer = combineReducers({
-   phone: phoneReducer,
-   tv: tvsReducer,
-   camera: camerasReducer,
- });
-
- const store = createStore(rootReducer);
-
- store.subscribe(() => {
-   const state = store.getState();
-   const { phone, tv, camera } = state;
-   console.log("Updated State:", state);
-   // Update the UI elements directly (Test)
-   document.getElementById("phones").innerHTML = `${phone.total} available`;
-   document.getElementById("tvs").innerHTML = `${tv.total} available`;
-   document.getElementById("cameras").innerHTML = `${camera.total} available`;
- });
-
- export { store, buyPhone, buyTV, buyCamera, phoneReducer };
```

AJOUTÉ **src/store/reducers.ts** +57 −0

```
@@ -1,0 +1,57 @@
+ // reducer
+
+ import { BUY_CAMERA, BUY_PHONE, BUY_TV } from "./actions";
+
+ // initial state
+ const initialStatePhone = {
+   total: 5,
+ };
+
+ const initialStateTV = {
+   total: 3,
+ };
+
+ const initialStateCamera = {
+   total: 8,
+ };
+
+ // (previousState, action) => newState
+ const phoneReducer = (state = initialStatePhone, action) => {
+   switch (action.type) {
+     case BUY_PHONE:
+       return {
+         ...state,
+         total: state.total - 1,
+       };
+     default:
+       return state;
+   }
```

```
+ };
+
+ // (previousState, action) => newState
+ const tvsReducer = (state = initialStateTV, action) => {
+   switch (action.type) {
+     case BUY_TV:
+       return {
+         ...state,
+         total: state.total - 1,
+       };
+     default:
+       return state;
+   }
+ };
+
+ // (previousState, action) => newState
+ const camerasReducer = (state = initialStateCamera, action) => {
+   switch (action.type) {
+     case BUY_CAMERA:
+       return {
+         ...state,
+         total: state.total - 1,
+       };
+     default:
+       return state;
+   }
+ };
+
+ export { phoneReducer, tvsReducer, camerasReducer };
```

AJOUTÉ **src/store/store.ts** +20 −0

```
@@ -1,0 +1,20 @@
+ import { combineReducers, createStore, applyMiddleware } from "redux";
+ import { camerasReducer, phoneReducer, tvsReducer } from "./reducers";
+
+ const rootReducer = combineReducers({
+   phone: phoneReducer,
+   tv: tvsReducer,
+   camera: camerasReducer,
+ });
+
+ // Simple logging middleware
+ const loggerMiddleware = (store) => (next) => (action) => {
+   console.log("Action:", action);
+   const result = next(action);
+   console.log("State after action:", store.getState());
+   return result;
+ };
+
+ const store = createStore(rootReducer, applyMiddleware(loggerMiddleware));
+
+ export { store };
```

Étape 3 → 4 etape-3-provider → etape-4-payload

Action avec payload : choisir la quantité de téléphones

MODIFIÉ **src/App.css** +13 −0

```
@@ -65,3 +65,16 @@
    font-weight: 600;
    color: white;
  }
+
+ .quantity {
+   margin: 0;
+ }
+
+ .input {
+   width: 60px;
+   padding: 4px;
+   font-size: 16px;
+   border: 1px solid #ccc;
+   border-radius: 4px;
+   text-align: center;
+ }
```

MODIFIÉ **src/pages/Home.tsx** +39 −2

```
@@ -5,15 +5,22 @@
  // @ts-ignore
  import cameraImg from "../assets/camera.png";

- import { store } from "../store/store";
  import { buyPhone, buyTV, buyCamera } from "../store/actions";
  import { useSelector, useDispatch } from "react-redux";
+ import { useState } from "react";

  function Home() {
+   const [quantity, setQuantity] = useState(1);
+
    const dispatch = useDispatch();
    const phone = useSelector((state: any) => state.phone);
    const camera = useSelector((state: any) => state.camera);
    const tv = useSelector((state: any) => state.tv);
+
+   const handleChange = (e) => {
+     const value = e.target.value;
+     setQuantity(value);
+   };

    return (
      <div>
@@ -24,10 +31,20 @@
            <div className="availability" id="phones">
              {phone.total} available
            </div>
+           <div className="quantity">
+             <input
+               className="input"
+               type="number"
+               min="1"
+               max={phone.total}
+               defaultValue={1}
+               onChange={handleChange}
+             />
```

```
          43 +           </div>
27        44             <button
28        45               className="button"
29        46               onClick={() => {
30          -              dispatch(buyPhone());
          47 +              dispatch(buyPhone(quantity));
31        48               }}
32        49             >
33        50               Buy
                 @@ -37,6 +54,16 @@
37        54             <img src={tvImg} alt="" width={100} height={100} />
38        55             <div className="availability" id="tvs">
39        56               {tv.total} available
          57 +           </div>
          58 +           <div className="quantity">
          59 +             <input
          60 +               className="input"
          61 +               type="number"
          62 +               min="1"
          63 +               max={phone.total}
          64 +               defaultValue={1}
          65 +               disabled
          66 +             />
40        67             </div>
41        68             <button
42        69               className="button"
                 @@ -51,6 +78,16 @@
51        78             <img src={cameraImg} alt="" width={100} height={100} />
52        79             <div className="availability" id="cameras">
53        80               {camera.total} available
          81 +           </div>
          82 +           <div className="quantity">
          83 +             <input
          84 +               className="input"
          85 +               type="number"
          86 +               min="1"
          87 +               max={phone.total}
          88 +               defaultValue={1}
          89 +               disabled
          90 +             />
54        91             </div>
55        92             <button
56        93               className="button"
```

MODIFIÉ **src/store/actions.ts** +2 −1

```
                 @@ -4,9 +4,10 @@
4         4    const BUY_TV = "BUY_TV";
5         5    const BUY_CAMERA = "BUY_CAMERA";
6         6
7           - const buyPhone = () => {
            7  + const buyPhone = (quantity) => {
8         8      return {
9         9        type: BUY_PHONE,
            10 +     payload: quantity,
10        11     };
11        12   };
12        13
```

MODIFIÉ **src/store/reducers.ts** +1 −1

```
                 @@ -21,7 +21,7 @@
21        21       case BUY_PHONE:
22        22         return {
```

```
23  23          ...state,
24      -       total: state.total - 1,
    24  +       total: state.total - action.payload,
25  25        };
26  26      default:
27  27        return state;
```

MODIFIÉ **src/store/store.ts** +5 −9

```
    @@ -1,4 +1,4 @@
1       - import { combineReducers, createStore, applyMiddleware } from "redux";
    1   + import { combineReducers, createStore } from "redux";
2   2   import { camerasReducer, phoneReducer, tvsReducer } from "./reducers";
3   3
4   4   const rootReducer = combineReducers({
    @@ -7,14 +7,10 @@
7   7     camera: camerasReducer,
8   8   });
9   9
10      - // Simple logging middleware
11      - const loggerMiddleware = (store) => (next) => (action) => {
12      -   console.log("Action:", action);
13      -   const result = next(action);
14      -   console.log("State after action:", store.getState());
15      -   return result;
16      - };
    10  + const store = createStore(rootReducer);
17  11
18      - const store = createStore(rootReducer, applyMiddleware(loggerMiddleware));
    12  + store.subscribe(() => {
    13  +   console.log("Updated State:", store.getState());
    14  + });
19  15
20  16  export { store };
```

Étape 4 → 5 etape-4-payload → etape-5-middleware

Middleware : redux-thunk + axios pour charger les commentaires (loading / success / error)

MODIFIÉ **package.json** +3 −1

```
@@ -10,10 +10,12 @@
    "preview": "vite preview"
  },
  "dependencies": {
+   "axios": "^1.13.2",
    "react": "^19.2.0",
    "react-dom": "^19.2.0",
    "react-redux": "^9.2.0",
-   "redux": "^5.0.1"
+   "redux": "^5.0.1",
+   "redux-thunk": "^3.1.0"
  },
  "devDependencies": {
    "@eslint/js": "^9.39.1",
```

MODIFIÉ **src/App.css** +35 −0

```
@@ -78,3 +78,38 @@
    border-radius: 4px;
    text-align: center;
  }
+
+ .page {
+   box-sizing: border-box;
+   place-items: center;
+ }
+
+ .comments-section {
+   padding: 12px;
+   width: 610px;
+   border: 0.5px solid mediumslateblue;
+   background-color: beige;
+   box-sizing: border-box;
+   border-radius: 4px;
+ }
+
+ .comment {
+   padding: 12px;
+   border: 1px solid #eee;
+   border-radius: 4px;
+   margin-bottom: 12px;
+   background-color: lightyellow;
+   border: none;
+ }
+
+ .comment-name {
+   font-weight: 600;
+   margin-bottom: 8px;
+   text-align: left;
+ }
+
+ .comment-body {
+   font-weight: 400;
+   white-space: pre-wrap;
+   text-align: left;
+ }
```

MODIFIÉ **src/pages/Home.tsx** +103 −70

```
@@ -7,7 +7,8 @@
 7  7
 8  8   import { buyPhone, buyTV, buyCamera } from "../store/actions";
 9  9   import { useSelector, useDispatch } from "react-redux";
10     - import { useState } from "react";
   10  + import { useEffect, useState } from "react";
   11  + import { apiCall } from "../store/comments/actions";
11 12
12 13   function Home() {
13 14     const [quantity, setQuantity] = useState(1);
@@ -16,87 +17,119 @@
16 17     const phone = useSelector((state: any) => state.phone);
17 18     const camera = useSelector((state: any) => state.camera);
18 19     const tv = useSelector((state: any) => state.tv);
   20  +   const comments = useSelector((state: any) => state.comments);
19 21
20 22     const handleChange = (e) => {
21 23       const value = e.target.value;
22 24       setQuantity(value);
23 25     };
24 26
   27  +   useEffect(() => {
   28  +     // Utiliser l'action thunk pour charger les commentaires
   29  +     dispatch(apiCall());
   30  +   }, []);
   31  +
25 32     return (
26 33       <div>
27 34         <h2>{"Welcome to our Application"}</h2>
28     -       <div className="container">
29     -         <div className="item">
30     -           <img src={phoneImg} alt="" width={100} height={100} />
31     -           <div className="availability" id="phones">
32     -             {phone.total} available
   35  +       <div className="page">
   36  +         <div className="container">
   37  +           <div className="item">
   38  +             <img src={phoneImg} alt="" width={100} height={100} />
   39  +             <div className="availability" id="phones">
   40  +               {phone.total} available
   41  +             </div>
   42  +             <div className="quantity">
   43  +               <input
   44  +                 className="input"
   45  +                 type="number"
   46  +                 min="1"
   47  +                 max={phone.total}
   48  +                 defaultValue={1}
   49  +                 onChange={handleChange}
   50  +               />
   51  +             </div>
   52  +             <button
   53  +               className="button"
   54  +               onClick={() => {
   55  +                 dispatch(buyPhone(quantity));
   56  +               }}
   57  +             >
   58  +               Buy
   59  +             </button>
33 60             </div>
34     -           <div className="quantity">
35     -             <input
```

```
36      -               className="input"
37      -               type="number"
38      -               min="1"
39      -               max={phone.total}
40      -               defaultValue={1}
41      -               onChange={handleChange}
42      -             />
        61  +           <div className="item">
        62  +             <img src={tvImg} alt="" width={100} height={100} />
        63  +             <div className="availability" id="tvs">
        64  +               {tv.total} available
        65  +             </div>
        66  +             <div className="quantity">
        67  +               <input
        68  +                 className="input"
        69  +                 type="number"
        70  +                 min="1"
        71  +                 max={phone.total}
        72  +                 defaultValue={1}
        73  +                 disabled
        74  +               />
        75  +             </div>
        76  +             <button
        77  +               className="button"
        78  +               onClick={() => {
        79  +                 dispatch(buyTV());
        80  +               }}
        81  +             >
        82  +               Buy
        83  +             </button>
43      84              </div>
44      -           <button
45      -             className="button"
46      -             onClick={() => {
47      -               dispatch(buyPhone(quantity));
48      -             }}
49      -           >
50      -             Buy
51      -           </button>
        85  +           <div className="item">
        86  +             <img src={cameraImg} alt="" width={100} height={100} />
        87  +             <div className="availability" id="cameras">
        88  +               {camera.total} available
        89  +             </div>
        90  +             <div className="quantity">
        91  +               <input
        92  +                 className="input"
        93  +                 type="number"
        94  +                 min="1"
        95  +                 max={phone.total}
        96  +                 defaultValue={1}
        97  +                 disabled
        98  +               />
        99  +             </div>
       100  +             <button
       101  +               className="button"
       102  +               onClick={() => {
       103  +                 dispatch(buyCamera());
       104  +               }}
       105  +             >
       106  +               Buy
       107  +             </button>
       108  +           </div>
52     109            </div>
53      -         <div className="item">
```

```
54  -           <img src={tvImg} alt="" width={100} height={100} />
55  -           <div className="availability" id="tvs">
56  -             {tv.total} available
57  -           </div>
58  -           <div className="quantity">
59  -             <input
60  -               className="input"
61  -               type="number"
62  -               min="1"
63  -               max={phone.total}
64  -               defaultValue={1}
65  -               disabled
66  -             />
67  -           </div>
68  -           <button
69  -             className="button"
70  -             onClick={() => {
71  -               dispatch(buyTV());
72  -             }}
73  -           >
74  -             Buy
75  -           </button>
76  -         </div>
77  -         <div className="item">
78  -           <img src={cameraImg} alt="" width={100} height={100} />
79  -           <div className="availability" id="cameras">
80  -             {camera.total} available
81  -           </div>
82  -           <div className="quantity">
83  -             <input
84  -               className="input"
85  -               type="number"
86  -               min="1"
87  -               max={phone.total}
88  -               defaultValue={1}
89  -               disabled
90  -             />
91  -           </div>
92  -           <button
93  -             className="button"
94  -             onClick={() => {
95  -               dispatch(buyCamera());
96  -             }}
97  -           >
98  -             Buy
99  -           </button>
110 +         <div className="comments-section">
111 +           {comments.isLoading ? (
112 +             <div>Chargement des commentaires...</div>
113 +           ) : comments.error ? (
114 +             <div>Erreur: {comments.error}</div>
115 +           ) : (
116 +             <div className="comments-list">
117 +               {comments.comments.length > 0 ? (
118 +                 comments.comments
119 +                   .slice(0, 10)
120 +                   .map((comment: any, index: number) => (
121 +                     <div key={comment.id || index} className="comment">
122 +                       <div className="comment-name">
123 +                         {comment.name || comment.email}
124 +                       </div>
125 +                       <div className="comment-body">{comment.body}</div>
126 +                     </div>
127 +                   ))
128 +               ) : (
```

```
                <div>Aucun commentaire disponible</div>
              )}
            </div>
          )}
        </div>
      </div>
    </div>
```

AJOUTÉ **src/store/comments/actions.ts** +40 −0

```
@@ -1,0 +1,40 @@
+ export const LOAD_COMMENTS = "LOAD_COMMENTS";
+ export const LOAD_COMMENTS_SUCCESS = "LOAD_COMMENTS_SUCCESS";
+ export const LOAD_COMMENTS_ERROR = "LOAD_COMMENTS_ERROR";
+
+ import axios from "axios";
+
+ const loadApiComments = () => {
+   return {
+     type: LOAD_COMMENTS,
+   };
+ };
+
+ const loadCommentsSuccess = (comments) => {
+   return {
+     type: LOAD_COMMENTS_SUCCESS,
+     payload: comments,
+   };
+ };
+
+ const loadCommentsError = (error) => {
+   return {
+     type: LOAD_COMMENTS_ERROR,
+     payload: error,
+   };
+ };
+
+ export const apiCall = () => {
+   return async (dispatch) => {
+     dispatch(loadApiComments());
+
+     try {
+       const response = await axios.get(
+         "https://jsonplaceholder.typicode.com/comments?postId=1"
+       );
+       dispatch(loadCommentsSuccess(response.data));
+     } catch (error) {
+       dispatch(loadCommentsError(error.message));
+     }
+   };
+ };
```

AJOUTÉ **src/store/comments/reducers.ts** +40 −0

```
@@ -1,0 +1,40 @@
+ import {
+   LOAD_COMMENTS,
+   LOAD_COMMENTS_ERROR,
+   LOAD_COMMENTS_SUCCESS,
+ } from "./actions";
+
+ const initialStateComments = {
+   isLoading: false,
+   comments: [],
+   error: "",
```

```
+ };
+
+ const commentReducer = (state = initialStateComments, action) => {
+   switch (action.type) {
+     case LOAD_COMMENTS:
+       return {
+         ...state,
+         isLoading: true,
+       };
+     case LOAD_COMMENTS_SUCCESS:
+       return {
+         ...state,
+         isLoading: false,
+         comments: action.payload,
+         error: "",
+       };
+     case LOAD_COMMENTS_ERROR:
+       return {
+         ...state,
+         isLoading: false,
+         comments: [],
+         error: action.payload,
+       };
+
+     default:
+       return state;
+   }
+ };
+
+ export default commentReducer;
```

MODIFIÉ **src/store/store.ts** +15 −2

```
@@ -1,13 +1,26 @@
- import { combineReducers, createStore } from "redux";
+ import { applyMiddleware, combineReducers, createStore, compose } from "redux";
  import { camerasReducer, phoneReducer, tvsReducer } from "./reducers";
+ import commentReducer from "./comments/reducers";
+ // Import nommé au lieu de default
+ import { thunk } from "redux-thunk";

  const rootReducer = combineReducers({
    phone: phoneReducer,
    tv: tvsReducer,
    camera: camerasReducer,
+   comments: commentReducer,
  });

- const store = createStore(rootReducer);
+ const store = createStore(rootReducer, applyMiddleware(thunk));
+
+ /*
+ const composeEnhancers = window.__REDUX_DEVTOOLS_EXTENSION_COMPOSE__ || compose;
+
+ const store = createStore(
+   rootReducer,
+   composeEnhancers(applyMiddleware(thunk))
+ );
+ */

  store.subscribe(() => {
    console.log("Updated State:", store.getState());
```

Étape 5 → 6 etape-5-middleware → etape-6-template

Redux Toolkit : configureStore, un premier createSlice (phone)

MODIFIÉ **package.json** +3 −3

```
@@ -10,12 +10,12 @@
10  10      "preview": "vite preview"
11  11    },
12  12    "dependencies": {
13      -   "axios": "^1.13.2",
    13  +   "@reduxjs/toolkit": "^2.11.2",
14  14      "react": "^19.2.0",
15  15      "react-dom": "^19.2.0",
16  16      "react-redux": "^9.2.0",
17      -   "redux": "^5.0.1",
18      -   "redux-thunk": "^3.1.0"
    17  +   "react-router-dom": "^7.11.0",
    18  +   "redux": "^5.0.1"
19  19    },
20  20    "devDependencies": {
21  21      "@eslint/js": "^9.39.1",
```

MODIFIÉ **src/App.css** +0 −35

```
@@ -78,38 +78,3 @@
78  78    border-radius: 4px;
79  79    text-align: center;
80  80  }
81      -
82      - .page {
83      -   box-sizing: border-box;
84      -   place-items: center;
85      - }
86      -
87      - .comments-section {
88      -   padding: 12px;
89      -   width: 610px;
90      -   border: 0.5px solid mediumslateblue;
91      -   background-color: beige;
92      -   box-sizing: border-box;
93      -   border-radius: 4px;
94      - }
95      -
96      - .comment {
97      -   padding: 12px;
98      -   border: 1px solid #eee;
99      -   border-radius: 4px;
100     -   margin-bottom: 12px;
101     -   background-color: lightyellow;
102     -   border: none;
103     - }
104     -
105     - .comment-name {
106     -   font-weight: 600;
107     -   margin-bottom: 8px;
108     -   text-align: left;
109     - }
110     -
111     - .comment-body {
112     -   font-weight: 400;
113     -   white-space: pre-wrap;
114     -   text-align: left;
```

```
115     - }
```

MODIFIÉ **src/App.jsx** +0 −1

```
        @@ -2,7 +2,6 @@
2   2     import Home from "./pages/Home.tsx";
3   3     import { Provider } from "react-redux";
4   4     import { store } from "./store/store.ts";
5       - 
6   5     function App() {
7   6       return (
8   7         <Provider store={store}>
```

MODIFIÉ **src/pages/Home.tsx** +23 −116

```
        @@ -5,131 +5,38 @@
5   5     // @ts-ignore
6   6     import cameraImg from "../assets/camera.png";
7   7     
8       - import { buyPhone, buyTV, buyCamera } from "../store/actions";
9       - import { useSelector, useDispatch } from "react-redux";
10      - import { useEffect, useState } from "react";
11      - import { apiCall } from "../store/comments/actions";
        8 + import { buyPhone, buyTv, buyCamera } from "../store/actions";
        9 + import { useDispatch, useSelector } from "react-redux";
12  10    
13  11    function Home() {
14      -   const [quantity, setQuantity] = useState(1);
15      - 
16  12      const dispatch = useDispatch();
17  13      const phone = useSelector((state: any) => state.phone);
        14 +   const tv = useSelector((state: any) => state.tv);
18  15      const camera = useSelector((state: any) => state.camera);
19      -   const tv = useSelector((state: any) => state.tv);
20      -   const comments = useSelector((state: any) => state.comments);
21      - 
22      -   const handleChange = (e) => {
23      -     const value = e.target.value;
24      -     setQuantity(value);
25      -   };
26      - 
27      -   useEffect(() => {
28      -     // Utiliser l'action thunk pour charger les commentaires
29      -     dispatch(apiCall());
30      -   }, []);
31      - 
32  16      return (
33  17        <div>
34  18          <h2>{"Welcome to our Application"}</h2>
35      -       <div className="page">
36      -         <div className="container">
37      -           <div className="item">
38      -             <img src={phoneImg} alt="" width={100} height={100} />
39      -             <div className="availability" id="phones">
40      -               {phone.total} available
41      -             </div>
42      -             <div className="quantity">
43      -               <input
44      -                 className="input"
45      -                 type="number"
46      -                 min="1"
47      -                 max={phone.total}
48      -                 defaultValue={1}
49      -                 onChange={handleChange}
50      -               />
```

```
51  -             </div>
52  -             <button
53  -               className="button"
54  -               onClick={() => {
55  -                 dispatch(buyPhone(quantity));
56  -               }}
57  -             >
58  -               Buy
59  -             </button>
60  -           </div>
61  -           <div className="item">
62  -             <img src={tvImg} alt="" width={100} height={100} />
63  -             <div className="availability" id="tvs">
64  -               {tv.total} available
65  -             </div>
66  -             <div className="quantity">
67  -               <input
68  -                 className="input"
69  -                 type="number"
70  -                 min="1"
71  -                 max={phone.total}
72  -                 defaultValue={1}
73  -                 disabled
74  -               />
75  -             </div>
76  -             <button
77  -               className="button"
78  -               onClick={() => {
79  -                 dispatch(buyTV());
80  -               }}
81  -             >
82  -               Buy
83  -             </button>
84  -           </div>
85  -           <div className="item">
86  -             <img src={cameraImg} alt="" width={100} height={100} />
87  -             <div className="availability" id="cameras">
88  -               {camera.total} available
89  -             </div>
90  -             <div className="quantity">
91  -               <input
92  -                 className="input"
93  -                 type="number"
94  -                 min="1"
95  -                 max={phone.total}
96  -                 defaultValue={1}
97  -                 disabled
98  -               />
99  -             </div>
100 -             <button
101 -               className="button"
102 -               onClick={() => {
103 -                 dispatch(buyCamera());
104 -               }}
105 -             >
106 -               Buy
107 -             </button>
108 -           </div>
19  +       <div className="container">
20  +         <div className="item">
21  +           <img src={phoneImg} alt="" width={100} height={100} />
22  +           <div className="availability">{phone.total} available</div>
23  +           <button className="button" onClick={() => dispatch(buyPhone(2))}>
24  +             Buy
25  +           </button>
```

```
109  26          </div>
110      -       <div className="comments-section">
111      -         {comments.isLoading ? (
112      -           <div>Chargement des commentaires...</div>
113      -         ) : comments.error ? (
114      -           <div>Erreur: {comments.error}</div>
115      -         ) : (
116      -           <div className="comments-list">
117      -             {comments.comments.length > 0 ? (
118      -               comments.comments
119      -                 .slice(0, 10)
120      -                 .map((comment: any, index: number) => (
121      -                   <div key={comment.id || index} className="comment">
122      -                     <div className="comment-name">
123      -                       {comment.name || comment.email}
124      -                     </div>
125      -                     <div className="comment-body">{comment.body}</div>
126      -                   </div>
127      -                 ))
128      -             ) : (
129      -               <div>Aucun commentaire disponible</div>
130      -             )}
131      -           </div>
132      -         )}
         27  +       <div className="item">
         28  +         <img src={tvImg} alt="" width={100} height={100} />
         29  +         <div className="availability">{tv.total} available</div>
         30  +         <button className="button" onClick={() => dispatch(buyTv())}>
         31  +           Buy
         32  +         </button>
         33  +       </div>{" "}
         34  +       <div className="item">
         35  +         <img src={cameraImg} alt="" width={100} height={100} />
         36  +         <div className="availability">{camera.total} available</div>
         37  +         <button className="button" onClick={() => dispatch(buyCamera())}>
         38  +           Buy
         39  +         </button>
133  40          </div>
134  41        </div>
135  42      </div>
```

MODIFIÉ **src/store/actions.ts** +6 −6

```
         @@ -1,8 +1,4 @@
1        - // Action Types
2        -
3    1   const BUY_PHONE = "BUY_PHONE";
4        - const BUY_TV = "BUY_TV";
5        - const BUY_CAMERA = "BUY_CAMERA";
6    2
7    3   const buyPhone = (quantity) => {
8    4     return {
         @@ -11,11 +7,15 @@
11   7     };
12   8   };
13   9
14       - const buyTV = () => {
         10 + const BUY_TV = "BUY_TV";
         11 +
         12 + const buyTv = () => {
15   13    return {
16   14      type: BUY_TV,
17   15    };
18   16  };
```

```
@@ -17,2 +17,3 @@
+
+ const BUY_CAMERA = "BUY_CAMERA";
 
   const buyCamera = () => {
     return {
@@ -23,4 +23,4 @@
     };
   };
 
- export { BUY_PHONE, BUY_TV, BUY_CAMERA, buyPhone, buyTV, buyCamera };
+ export { BUY_PHONE, buyPhone, BUY_TV, buyTv, BUY_CAMERA, buyCamera };
```

SUPPRIMÉ **src/store/comments/actions.ts** +0 −40

```
@@ -1,40 +1,0 @@
- export const LOAD_COMMENTS = "LOAD_COMMENTS";
- export const LOAD_COMMENTS_SUCCESS = "LOAD_COMMENTS_SUCCESS";
- export const LOAD_COMMENTS_ERROR = "LOAD_COMMENTS_ERROR";
-
- import axios from "axios";
-
- const loadApiComments = () => {
-   return {
-     type: LOAD_COMMENTS,
-   };
- };
-
- const loadCommentsSuccess = (comments) => {
-   return {
-     type: LOAD_COMMENTS_SUCCESS,
-     payload: comments,
-   };
- };
-
- const loadCommentsError = (error) => {
-   return {
-     type: LOAD_COMMENTS_ERROR,
-     payload: error,
-   };
- };
-
- export const apiCall = () => {
-   return async (dispatch) => {
-     dispatch(loadApiComments());
-
-     try {
-       const response = await axios.get(
-         "https://jsonplaceholder.typicode.com/comments?postId=1"
-       );
-       dispatch(loadCommentsSuccess(response.data));
-     } catch (error) {
-       dispatch(loadCommentsError(error.message));
-     }
-   };
- };
```

SUPPRIMÉ **src/store/comments/reducers.ts** +0 −40

```
@@ -1,40 +1,0 @@
- import {
-   LOAD_COMMENTS,
-   LOAD_COMMENTS_ERROR,
-   LOAD_COMMENTS_SUCCESS,
- } from "./actions";
```

```
-
- const initialStateComments = {
-   isLoading: false,
-   comments: [],
-   error: "",
- };
-
- const commentReducer = (state = initialStateComments, action) => {
-   switch (action.type) {
-     case LOAD_COMMENTS:
-       return {
-         ...state,
-         isLoading: true,
-       };
-     case LOAD_COMMENTS_SUCCESS:
-       return {
-         ...state,
-         isLoading: false,
-         comments: action.payload,
-         error: "",
-       };
-     case LOAD_COMMENTS_ERROR:
-       return {
-         ...state,
-         isLoading: false,
-         comments: [],
-         error: action.payload,
-       };
-
-     default:
-       return state;
-   }
- };
-
- export default commentReducer;
```

AJOUTÉ **src/store/phoneSlice.ts** +16 −0

```
@@ -1,0 +1,16 @@
+ import { createSlice } from "@reduxjs/toolkit";
+
+ const phoneSlice = createSlice({
+   name: "phone",
+   initialState: {
+     total: 10,
+   },
+   reducers: {
+     buyPhone: (state, action) => {
+       state.total -= action.payload || 1;
+     },
+   },
+ });
+
+ export const { buyPhone } = phoneSlice.actions;
+ export const phoneReducer = phoneSlice.reducer;
```

MODIFIÉ **src/store/reducers.ts** +14 −21

```
@@ -1,22 +1,10 @@
- // reducer
+ import { BUY_PHONE, BUY_TV, BUY_CAMERA } from "./actions";

- import { BUY_CAMERA, BUY_PHONE, BUY_TV } from "./actions";
-
- // initial state
```

```
6       - const initialStatePhone = {
7       -   total: 5,
        3 + const phoneInitialState = {
        4 +   total: 10,
8       5   };
9       6
10      - const initialStateTV = {
11      -   total: 3,
12      - };
13      -
14      - const initialStateCamera = {
15      -   total: 8,
16      - };
17      -
18      - // (previousState, action) => newState
19      - const phoneReducer = (state = initialStatePhone, action) => {
        7 + const phoneReducer = (state = phoneInitialState, action) => {
20      8     switch (action.type) {
21      9       case BUY_PHONE:
22      10        return {
@@ -27,9 +15,11 @@
27      15        return state;
28      16    }
29      17  };
        18 + const tvInitialState = {
        19 +   total: 8,
        20 + };
30      21
31      - // (previousState, action) => newState
32      - const tvsReducer = (state = initialStateTV, action) => {
        22 + const tvReducer = (state = tvInitialState, action) => {
33      23    switch (action.type) {
34      24      case BUY_TV:
35      25        return {
@@ -41,8 +31,11 @@
41      31    }
42      32  };
43      33
44      - // (previousState, action) => newState
45      - const camerasReducer = (state = initialStateCamera, action) => {
        34 + const cameraInitialState = {
        35 +   total: 5,
        36 + };
        37 +
        38 + const cameraReducer = (state = cameraInitialState, action) => {
46      39    switch (action.type) {
47      40      case BUY_CAMERA:
48      41        return {
@@ -54,4 +47,4 @@
54      47    }
55      48  };
56      49
57      - export { phoneReducer, tvsReducer, camerasReducer };
        50 + export { phoneReducer, tvReducer, cameraReducer };
```

MODIFIÉ **src/store/store.ts** +7 −27

```
@@ -1,29 +1,9 @@
1       - import { applyMiddleware, combineReducers, createStore, compose } from "redux";
2       - import { camerasReducer, phoneReducer, tvsReducer } from "./reducers";
3       - import commentReducer from "./comments/reducers";
4       - // Import nommé au lieu de default
5       - import { thunk } from "redux-thunk";
        1 + import { configureStore } from "@reduxjs/toolkit";
```

```
+ import { phoneReducer } from "./phoneSlice";

- const rootReducer = combineReducers({
-   phone: phoneReducer,
-   tv: tvsReducer,
-   camera: camerasReducer,
-   comments: commentReducer,
+ export const store = configureStore({
+   reducer: {
+     phone: phoneReducer,
+   },
+   devTools: true,
  });
-
- const store = createStore(rootReducer, applyMiddleware(thunk));
-
- /*
- const composeEnhancers = window.__REDUX_DEVTOOLS_EXTENSION_COMPOSE__ || compose;
-
- const store = createStore(
-   rootReducer,
-   composeEnhancers(applyMiddleware(thunk))
- );
- */
-
- store.subscribe(() => {
-   console.log("Updated State:", store.getState());
- });
-
- export { store };
```

Étape 6 → 7 etape-6-template → etape-7-redux-toolkit

Les 3 slices (phone, tv, camera), extraReducers (promo TV), RTK Query (createApi)

MODIFIÉ `package.json` +0 −1

```
@@ -14,7 +14,6 @@
      "react": "^19.2.0",
      "react-dom": "^19.2.0",
      "react-redux": "^9.2.0",
-     "react-router-dom": "^7.11.0",
      "redux": "^5.0.1"
    },
    "devDependencies": {
```

MODIFIÉ `src/App.css` +0 −13

```
@@ -65,16 +65,3 @@
    font-weight: 600;
    color: white;
  }
-
- .quantity {
-   margin: 0;
- }
-
- .input {
-   width: 60px;
-   padding: 4px;
-   font-size: 16px;
-   border: 1px solid #ccc;
-   border-radius: 4px;
-   text-align: center;
- }
```

MODIFIÉ `src/App.jsx` +1 −0

```
@@ -2,6 +2,7 @@
  import Home from "./pages/Home.tsx";
  import { Provider } from "react-redux";
  import { store } from "./store/store.ts";
+
  function App() {
    return (
      <Provider store={store}>
```

AJOUTÉ `src/apis/index.ts` +70 −0

```
@@ -1,0 +1,70 @@
+ // RTK Query (Redux Toolkit Query) : on crée une "API slice"
+ // Objectif : faire des appels HTTP + gérer cache / loading / error automatiquement
+
+ import { createApi, fetchBaseQuery } from "@reduxjs/toolkit/query/react";
+
+ /**
+  * commentsApi = "API slice"
+  * Elle contient :
+  * - un reducer (pour stocker cache + status dans Redux)
+  * - un middleware (pour déclencher/optimiser les requêtes)
+  * - des endpoints (GET/POST/...)
+  * - des hooks React auto-générés (useGetCommentsQuery, etc.)
+  */
+ export const commentsApi = createApi({
```

```
+   /**
+    * reducerPath = le nom de la clé dans le store Redux
+    * Exemple : state.commentsApi => contiendra le cache, loading, erreurs, etc.
+    *
+    * Dans le store on fera :
+    * reducer: { [commentsApi.reducerPath]: commentsApi.reducer }
+    */
+   reducerPath: "commentsApi",
+
+   /**
+    * baseQuery = le "client HTTP" utilisé par RTK Query
+    * fetchBaseQuery est une petite couche au-dessus de fetch (pas besoin d'axios)
+    *
+    * baseUrl = URL de base, les endpoints ajouteront juste le chemin
+    */
+   baseQuery: fetchBaseQuery({
+     baseUrl: "https://jsonplaceholder.typicode.com",
+   }),
+
+   /**
+    * endpoints = on déclare ici toutes les routes API
+    * builder permet de créer :
+    * - builder.query(...)  => GET (lecture) + cache
+    * - builder.mutation(...) => POST/PUT/DELETE (écriture)
+    */
+   endpoints: (builder) => ({
+     /**
+      * getComments = nom de l'endpoint
+      * builder.query => c'est un endpoint de lecture (GET)
+      */
+     getComments: builder.query({
+       /**
+        * query = fonction qui construit l'URL de la requête
+        * postId est un paramètre (optionnel) avec une valeur par défaut = 1
+        *
+        * Si postId = 1 => GET /comments?postId=1
+        * Si postId = 5 => GET /comments?postId=5
+        *
+        * RTK Query gère automatiquement :
+        * - loading / success / error
+        * - cache + dedupe (évite des appels identiques)
+        */
+       query: (postId = 1) => `/comments?postId=${postId}`,
+     }),
+   }),
+ });
+
+ /**
+  * RTK Query génère automatiquement des hooks React
+  * Le nom est basé sur l'endpoint :
+  * getComments => useGetCommentsQuery
+  *
+  * Utilisation :
+  * const { data, isLoading, error } = useGetCommentsQuery(1);
+  */
+ export const { useGetCommentsQuery } = commentsApi;
```

MODIFIÉ **src/pages/Home.tsx** +38 −10

```
@@ -5,36 +5,64 @@
  // @ts-ignore
  import cameraImg from "../assets/camera.png";

- import { buyPhone, buyTv, buyCamera } from "../store/actions";
- import { useDispatch, useSelector } from "react-redux";
```

```
+ import { buyPhone } from "../store/phoneSlice";
+ import { buyTv } from "../store/tvSlice";
+ import { buyCamera } from "../store/cameraSlice";
+
+ import { useSelector, useDispatch } from "react-redux";
+ import { useGetCommentsQuery } from "../apis";

  function Home() {
    const dispatch = useDispatch();
    const phone = useSelector((state: any) => state.phone);
+   const camera = useSelector((state: any) => state.camera);
    const tv = useSelector((state: any) => state.tv);
-   const camera = useSelector((state: any) => state.camera);
+   const { data, isLoading, error } = useGetCommentsQuery(1);
+   console.log({ data, isLoading, error });
+
    return (
      <div>
        <h2>{"Welcome to our Application"}</h2>
        <div className="container">
          <div className="item">
            <img src={phoneImg} alt="" width={100} height={100} />
-           <div className="availability">{phone.total} available</div>
-           <button className="button" onClick={() => dispatch(buyPhone(2))}>
+           <div className="availability" id="phones">
+             {phone.total} available
+           </div>
+           <button
+             className="button"
+             onClick={() => {
+               dispatch(buyPhone());
+             }}
+           >
              Buy
            </button>
          </div>
          <div className="item">
            <img src={tvImg} alt="" width={100} height={100} />
-           <div className="availability">{tv.total} available</div>
-           <button className="button" onClick={() => dispatch(buyTv())}>
+           <div className="availability" id="tvs">
+             {tv.total} available
+           </div>
+           <button
+             className="button"
+             onClick={() => {
+               dispatch(buyTv());
+             }}
+           >
              Buy
            </button>
-         </div>{" "}
+         </div>
          <div className="item">
            <img src={cameraImg} alt="" width={100} height={100} />
-           <div className="availability">{camera.total} available</div>
-           <button className="button" onClick={() => dispatch(buyCamera())}>
+           <div className="availability" id="cameras">
+             {camera.total} available
+           </div>
+           <button
+             className="button"
+             onClick={() => {
+               dispatch(buyCamera());
+             }}
```

```
      65  +           >
38    66                Buy
39    67              </button>
40    68            </div>
```

SUPPRIMÉ **src/store/actions.ts** +0 −26

```
@@ -1,26 +1,0 @@
1     -  const BUY_PHONE = "BUY_PHONE";
2     -
3     -  const buyPhone = (quantity) => {
4     -    return {
5     -      type: BUY_PHONE,
6     -      payload: quantity,
7     -    };
8     -  };
9     -
10    -  const BUY_TV = "BUY_TV";
11    -
12    -  const buyTv = () => {
13    -    return {
14    -      type: BUY_TV,
15    -    };
16    -  };
17    -
18    -  const BUY_CAMERA = "BUY_CAMERA";
19    -
20    -  const buyCamera = () => {
21    -    return {
22    -      type: BUY_CAMERA,
23    -    };
24    -  };
25    -
26    -  export { BUY_PHONE, buyPhone, BUY_TV, buyTv, BUY_CAMERA, buyCamera };
```

AJOUTÉ **src/store/cameraSlice.ts** +17 −0

```
@@ -1,0 +1,17 @@
1     +  // features/stock/cameraSlice.ts
2     +  import { createSlice } from "@reduxjs/toolkit";
3     +
4     +  const initialState = { total: 8 };
5     +
6     +  const cameraSlice = createSlice({
7     +    name: "camera",
8     +    initialState,
9     +    reducers: {
10    +      buyCamera(state) {
11    +        state.total -= 1;
12    +      },
13    +    },
14    +  });
15    +
16    +  export const { buyCamera } = cameraSlice.actions;
17    +  export default cameraSlice.reducer;
```

MODIFIÉ **src/store/phoneSlice.ts** +23 −7

```
@@ -1,16 +1,32 @@
1     -  import { createSlice } from "@reduxjs/toolkit";
1     +  // features/stock/phoneSlice.ts
2     +  import { createSlice, PayloadAction } from "@reduxjs/toolkit";
3     +  // import { buyTv, buyTvPromotion } from "./tvSlice";
4     +
5     +  const initialState = { total: 5 };
```

```
 2   6
 3   7    const phoneSlice = createSlice({
 4   8      name: "phone",
 5       -  initialState: {
 6       -    total: 10,
 7       -  },
         9  +  initialState,
 8   10     reducers: {
 9       -    buyPhone: (state, action) => {
10       -      state.total -= action.payload || 1;
        11  +    buyPhone(state) {
        12  +      // action.payload can be used for more complex logic
        13  +      state.total -= 1;
11   14       },
12   15     },
        16  +  /*extraReducers: (builder) => {
        17  +    builder.addCase(buyTv, (state) => {
        18  +      state.total -= 1; // si achat TV => +1 phone
        19  +    });
        20  +  },*/
        21  +  /*
        22  +    extraReducers: (builder) => {
        23  +    builder.addCase(buyTvPromotion, (state, action: PayloadAction<string>) => {
        24  +      if (action.payload === "2025XMAS") {
        25  +        state.total -= 1;
        26  +      }
        27  +    });
        28  +  }, */
13   29   });
14   30
15   31   export const { buyPhone } = phoneSlice.actions;
16       - export const phoneReducer = phoneSlice.reducer;
        32 + export default phoneSlice.reducer;
```

SUPPRIMÉ **src/store/reducers.ts** +0 −50

```
@@ -1,50 +1,0 @@
1   - import { BUY_PHONE, BUY_TV, BUY_CAMERA } from "./actions";
2   -
3   - const phoneInitialState = {
4   -   total: 10,
5   - };
6   -
7   - const phoneReducer = (state = phoneInitialState, action) => {
8   -   switch (action.type) {
9   -     case BUY_PHONE:
10  -       return {
11  -         ...state,
12  -         total: state.total - action.payload,
13  -       };
14  -     default:
15  -       return state;
16  -   }
17  - };
18  - const tvInitialState = {
19  -   total: 8,
20  - };
21  -
22  - const tvReducer = (state = tvInitialState, action) => {
23  -   switch (action.type) {
24  -     case BUY_TV:
25  -       return {
26  -         ...state,
27  -         total: state.total - 1,
28  -       };
```

```
-     default:
-       return state;
-   }
- };
-
- const cameraInitialState = {
-   total: 5,
- };
-
- const cameraReducer = (state = cameraInitialState, action) => {
-   switch (action.type) {
-     case BUY_CAMERA:
-       return {
-         ...state,
-         total: state.total - 1,
-       };
-     default:
-       return state;
-   }
- };
-
- export { phoneReducer, tvReducer, cameraReducer };
```

MODIFIÉ `src/store/store.ts` +11 −2

```
@@ -1,9 +1,18 @@
+ // store.ts
  import { configureStore } from "@reduxjs/toolkit";
- import { phoneReducer } from "./phoneSlice";
+ import phoneReducer from "./phoneSlice";
+ import tvReducer from "./tvSlice";
+ import cameraReducer from "./cameraSlice";
+ import { commentsApi } from "../apis";

  export const store = configureStore({
    reducer: {
      phone: phoneReducer,
+     tv: tvReducer,
+     camera: cameraReducer,
+     commentsApi: commentsApi.reducer,
    },
-   devTools: true,
+   middleware: (getDefaultMiddleware) =>
+     getDefaultMiddleware().concat(commentsApi.middleware),
+   devTools: true, // optionnel (par défaut true en dev)
  });
```

AJOUTÉ `src/store/tvSlice.ts` +20 −0

```
@@ -1,0 +1,20 @@
+ // features/stock/tvSlice.ts
+ import { createSlice, PayloadAction } from "@reduxjs/toolkit";
+
+ const initialState = { total: 3 };
+
+ const tvSlice = createSlice({
+   name: "tv",
+   initialState,
+   reducers: {
+     buyTv(state) {
+       state.total -= 1;
+     },
+     /*buyTvPromotion(state, action: PayloadAction<string>) {
+       state.total -= 1;
+     }/*, */
```

```
16 +   },
17 + });
18 +
19 + export const { buyTv } = tvSlice.actions;
20 + export default tvSlice.reducer;
```